

TESTs - Lesson 07 - Coding - Ballot.sol

When setting up a new Hardhat project by copying from an old one, which folder should you NOT copy to save time?

- A contracts
- B node_modules
- C cache
- D test

In the Ballot.sol contract, who is automatically assigned as the 'chairperson'?

- A The first person to vote.
- B The person who deploys the smart contract.
- C A randomly selected user.

Is it possible to directly import a Solidity (.sol) file into a TypeScript (.ts) test file?

- A Yes, always.
- B No, they are totally different worlds that don't communicate directly.
- C Only if the Solidity file is very small.

When interacting with a Solidity smart contract function from TypeScript, why is it necessary to use the `await` keyword for functions that return a Promise?

- A To force the TypeScript compiler to automatically convert the Solidity data type.
- B To ensure the Solidity function executes synchronously on the blockchain.
- C To pause execution until the Promise resolves, allowing access to the actual returned value.
- D To prevent gas fees from being charged incorrectly.

In the context of testing smart contracts with Hardhat and Viem, what is the main benefit of using `loadFixture(deployContract)`?

- A It automatically deploys multiple smart contracts simultaneously.
- B It allows for direct import of Solidity files into TypeScript tests.
- C It restores the blockchain to a snapshot state before each test, avoiding redundant deployment code.**
- D It enables remote debugging of smart contracts on public testnets.

What is the purpose of using `async/await` in TypeScript tests when interacting with smart contracts?

- A To make the code run faster.
- B To handle synchronous operations.
- C To manage asynchronous operations by waiting for promises to resolve before proceeding.**
- D To prevent the compiler from optimizing the code.

What are the tools and languages we used in this lesson for smart contract development and testing?

- A Python, Django, PostgreSQL
- B JavaScript, React, Node.js
- C TypeScript, Hardhat, Viem, Mocha, Chai, Solidity**
- D C++, Unreal Engine, SQL

What is the role of `loadFixture` in Hardhat testing?

- A To deploy multiple contracts simultaneously without a snapshot.
- B To create a temporary blockchain network for each test.
- C To set up the desired state of the network once, take a snapshot, and revert to it for subsequent tests to avoid repetition.**
- D To automatically generate test cases for all smart contract functions.

In Solidity, what is the significance of the `bytes32` type as discussed in the context of proposals?

- A It represents a large integer.
- B It stores short names (up to 32 bytes).**
- C It is used for cryptographic hashes only.